IA05: API Contract and Failure-Response Walkthrough

Identity: jpkanquandah, team Intellectuals. Repositories: https://github.com/Dept-of-Comp-Sci-University-of-Ghana/DE-IND-jpkanquandah and https://github.com/Dept-of-Comp-Sci-University-of-Ghana/dcit208-client-project-2026-team-engineering-repository-intellectuals. Story: customer enquires about a service. Issue: (link in the Sakai portfolio). Date: 23 October 2026.

Honest note: the website has no HTTP API. The "caller and provider agreement" for this feature is the WhatsApp link builder, a function inside the site that the Enquire button calls. The task allows a non-HTTP equivalent, so I define that contract. I use the descriptive name buildWhatsAppLink for it; the site code may name or inline this logic differently, which does not change the contract.

Purpose and contract

Item	Definition
Role, goal, value	As a customer, I want Enquire to open WhatsApp with the right number and a message about my service, so I do not type it all
Caller	The Enquire button on the Services section (and the "Not sure?" button)
Provider	The link builder function in the front-end code
NFR or constraint	Always use the approved number (NFR-12); store nothing (NFR-24)
Operation	buildWhatsAppLink(serviceId) returns a link string (non-HTTP equivalent of GET /enquiry-link)
Final link form	https://wa.me/<approved number, digits only>?text=<URL-encoded message> (example: https://wa.me/233XXXXXXXXX?text=Hello%2C%20I%20would%20like%20to%20ask%20about%20Mobile%20Crane%20Hire.)
Path or query parameters	serviceId (text, required for a service enquiry; omitted for the general message)
Headers and authentication	None. The function runs in the visitor's browser.
Authorization	None for visitors. Changing the number or service list is controlled by GitHub review and CODEOWNERS.

Request fields:

Field	Type	Required	Meaning
serviceId	text	Optional	Id of a service in the site content, for example “mobile-crane-hire” (synthetic example)
Message template (config)	text	Required	The fixed sentence with a place for the service name
Number (config)	text of digits	Required	Approved WhatsApp number, international format, no plus or spaces

Validation rules (5)

#	Rule	Kind	Checked by
1	serviceId must be text	Type	link builder
2	serviceId must match a service in the content list	Business	link builder
3	The number must be digits only, 10 to 15 digits	Format	config check or test
4	The message must be URL-encoded so spaces and symbols do not break the link	Format	link builder, tests TC-08
5	The number used must be the approved number	Business	tests TC-06 and TC-19

Success response

Outcome: a link is returned and the browser opens WhatsApp. Synthetic example response:

```
{
  "ok": true,
  "link": "https://wa.me/233000000000?text=Hello%2C%20I%20would%20like%20to%20ask%20about%20Mobile%20Crane%20Hire.",
  "service": "Mobile Crane Hire"
}
```

Fields: ok says it worked; link is what the button opens; service is the name used in the message. Next display: WhatsApp opens with the message ready to edit. (The real function may return only the link string. The JSON is how I describe the contract.)

Error cases (5)

Case	Outcome	Error code	User-facing message	Logging
Invalid input: serviceId is not text	No link; fall back to the general	ERR_INVALID_SERVI	“Please ask the owner about this service.”	Console warning in development

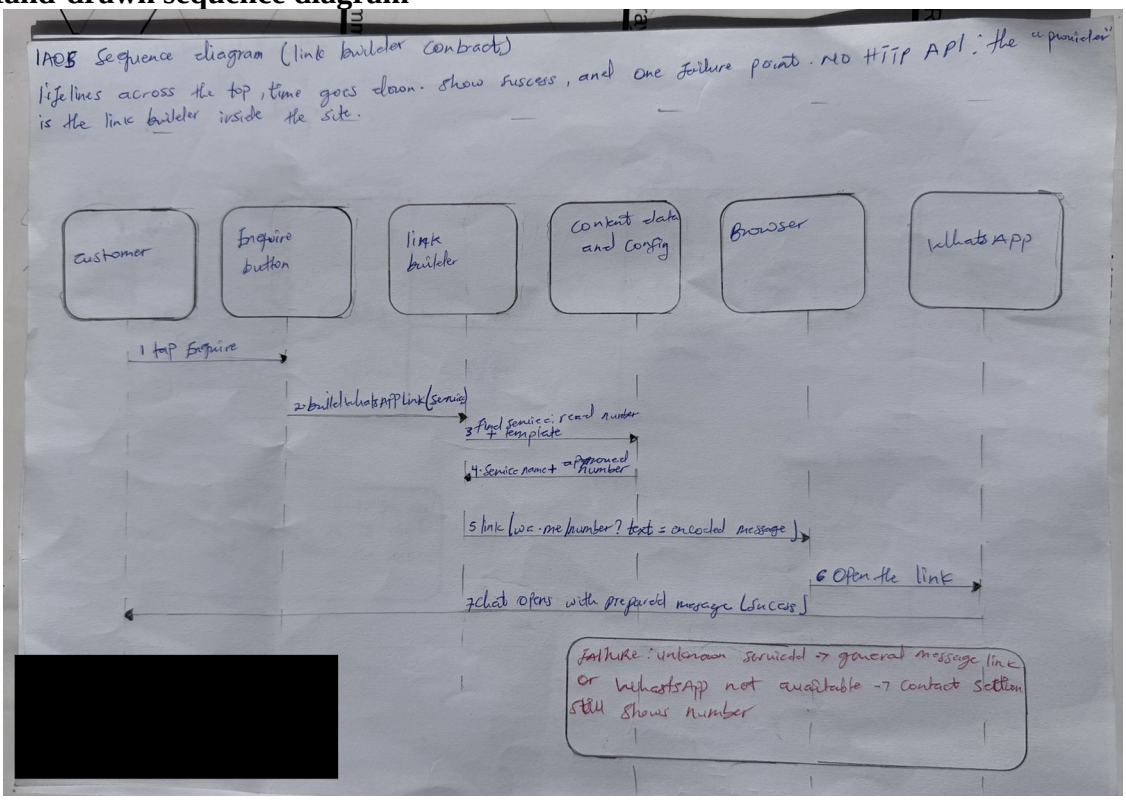
Case	Outcome	Error code	User-facing message	Logging
	message	CE_ID		
Authorization: someone tries to change the number without review	Change is blocked by branch protection	ERR_REVIEW_REQUIRED	Not visible to visitors; shown on the pull request	GitHub audit trail
Conflict: serviceId matches two services (duplicate id)	Build or test fails	ERR_DUPLICATE_SERVICE	Not shown to visitors; the change cannot be merged	CI log
External service failure: WhatsApp cannot open	Phone shows the WhatsApp page or an error	ERR_WHATSAPP_UNAVAILABLE	Contact section still shows the number	None on the site (nothing is collected)
Missing configuration: number is empty	Build or test fails	ERR_NUMBER_MISSING	Not shown to visitors	CI log

Status of each error case: ERR_REVIEW_REQUIRED is enforced today by GitHub branch protection and CODEOWNERS. The other codes (ERR_INVALID_SERVICE_ID, ERR_DUPLICATE_SERVICE, ERR_NUMBER_MISSING) are the names I give to checks that are planned in the link builder and CI tests, not confirmed in the code. ERR_WHATSAPP_UNAVAILABLE is a behaviour of the phone and WhatsApp, not of our code.

Duplicates and retries

If a customer taps Enquire twice, two chats may open, but nothing is stored, so there is no duplicate record. The function is idempotent (the same input always gives the same link). A retry is **safe** because it only builds a link. There is no unique constraint or request id because there is no data. The one unsafe retry is the customer sending the same WhatsApp message several times, which is outside our system.

Hand-drawn sequence diagram



IA05 handwritten sequence diagram

IA05 handwritten sequence diagram

AI note on this drawing: I drew this by hand myself. The layout was based on a reference sketch that Claude suggested, and I can explain every box and arrow in my own words.

Contract tests (6)

#	Test	Input	Expected outcome and response
1	Valid request	serviceId = existing service	ok true; link starts with wa.me and the approved number; message contains the service name
2	Missing or invalid field	serviceId = number 123	ERR_INVALID_SERVICE_ID; general message link
3	Unauthorized change	Pull request that edits the number without a reviewer	Merge blocked (process test: branch protection and CODEOWNERS)
4	Conflict or duplicate	Two services with the same id	CI fails with ERR_DUPLICATE_SERVICE
5	Dependency failure	Open the link on a phone without WhatsApp	Phone shows the WhatsApp page; Contact section still shows the number
6	Response	Valid request	Result contains ok, link and service with the

#	Test	Input	Expected outcome and response
	shape		right types

Trade-off, video and files

- **Trade-off and open question:** a function contract is simple but it cannot protect us if someone edits the number in the source, so we rely on review. Open question: should the number be stored only in one configuration file with its own automated check, or is review enough for a small project?
- Video (3 to 4 minutes): https://youtu.be/bMEDxWQgDD4.
- Files: IA05/API_CONTRACT.md, ERROR_MODEL.md, TEST_PLAN.md, examples/request.json, examples/success-response.json, examples/error-response.json, evidence/sequence-diagram.pdf, AI_DISCLOSURE.md.
- **AI used:** Yes. Claude (Anthropic), 23 October 2026: drafted the contract tables from my team’s documents. I changed names and examples to use synthetic values only, and marked planned checks as planned. I have not verified the function against the source code, which is why the contract is described as a design.